ENRIQUE ALVERDE

enrique.alverde@gmail.com | +1 (707)-787-7674 | kique.dev

EDUCATION

Columbia University — B.S. Computer Science **Aug 2024 – May 2026**
University of Redlands — B.A. Organized Thought through Fundamental Systems **Aug 2021 – May 2024**
Specialization: Computer Science, Physics, & Philosophy

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, TypeScript, Python, HTML, CSS, PHP
Frameworks & Tools: React Native, Expo, Node.js, Supabase, PostgreSQL, TanStack Query, Zustand, Zod, Git, Figma
Courses: Algorithms & Complexity, Data Structures, Computer Networks, Compilers, Databases, Cryptography, AI, Advanced C, C++, 3D UI/VR Development, Quantum Computing, Parallel Optimization, Web & Mobile Development

PROJECTS

Divvy — Collaborative Expense Splitting App **2024 – Present**
React Native, Expo, Supabase, PostgreSQL, TanStack Query, Zustand, Zod
- Designed a full PostgreSQL schema with row-level security policies and security definer functions
- Implemented database triggers for status propagation and a mutual friendship system across user accounts
- Scaffolded a React Native frontend with Expo Router file-based navigation

Receipt Scanner — Node.js CLI Tool **2026**
Node.js, Gemini API, JavaScript
- Built a CLI tool that scans a receipt photo via Gemini Vision API and writes validated JSON output to disk
- Implemented floating point rounding, async/await error handling, and environment variable management for API key security

VR Chapel — Immersive 3D Experience **2026**
Unity 6, C#, XR Interaction Toolkit, Meta Quest 2
- Built an immersive VR chapel environment for Meta Quest 2 using Unity 6 and the XR Interaction Toolkit
- Modeled architectural geometry in ProBuilder and implemented interactive object behaviors in C#

Blockchain P2P Network — Distributed Ledger System **2025**
Python, TCP Sockets, Distributed Systems
- Built a peer-to-peer blockchain network in Python with a tracker server for peer discovery
- Implemented proof-of-work mining, transaction pool management, and block broadcasting
- Built chain validation with longest-chain consensus across distributed peer nodes
- Designed the protocol for block propagation, chain sync, and graceful peer departure

PROFESSIONAL EXPERIENCE

The Remedy Project — Backend Developer **2024 – Present**
Supabase, PostgreSQL, Retool
- Designed and maintained a production PostgreSQL schema for a nonprofit supporting incarcerated individuals
- Configured row-level security policies, security definer functions, and append-only audit history tables
- Connected backend to a Retool frontend used by active case managers

University of Redlands — Web Assistant **Aug 2023 – Apr 2024**
- Coordinated a university website redesign across multiple departments
- Translated departmental goals into design requirements and applied consistent standards site-wide

City of Petaluma — Website Redesign **Summer 2023**
Proud City, CSS, HTML
- Redesigned the City of Petaluma's public website including the landing page, primary navigation, and overall site architecture
- Prioritized common resident use cases to improve information accessibility and navigation clarity